

Finitely generated vector spaces

Defⁿ W is a finitely generated k -vector space if the following equiv. conditions hold:

(1) $\exists$ finite set
 $X \subseteq W$

s.t. $\langle X \rangle_k = W$

(2) $\exists$ a surjective k -linear transformation

$$T: k^n \longrightarrow W$$

for some $n \geq 1$.

$$\boxed{(1) \Rightarrow (2)}$$

$$X = \{w_1, w_2, \dots, w_n\} \subseteq W$$

$$T: k^n \longrightarrow W$$

$$e_i \longmapsto w_i$$

$$\text{im } T = \langle \{w_1, \dots, w_n\} \rangle_k$$

$$= \langle X \rangle_k$$

$$= W$$

$\Rightarrow T$ is surjective

$$\boxed{(2) \Rightarrow (1)}$$

$$T: k^n \longrightarrow W$$

$$e_i \longmapsto w_i$$

surjective

$$W = \text{im } T = \langle \{w_1, \dots, w_n\} \rangle_k$$

$\Rightarrow$ Can take

$$X = \{w_1, \dots, w_n\}$$



We can add a third equivalent

Statement:

$$\begin{array}{ccc}
 \textcircled{3} & k^n & \\
 & \downarrow & \searrow T: \text{im } T = W \\
 & k^n / \ker T & \xrightarrow[\cong]{T} W
 \end{array}$$

i.e. W is isomorphic to
 a quotient vector space of
 the form k^n / U where
 $U \subseteq k^n$ is a k -subspace.

k -Subspaces of k^n

Proposition: (1) Every k -subspace of k^n is isomorphic to k^m for $m \leq n$

(2) If $U \leq k^n$ is a k -subspace isomorphic to k^m , then

$$k^n / U \cong k^{n-m}.$$

Corollary: (1) If $k^n \cong k^m$, then $n = m$

(2) W : f.s. k -vector space

Then $W \cong k^d$, for a unique $d \geq 0$.

Defⁿ If W is a f.g k -vector space,
the dimension of W is the
unique $d \in \mathbb{Z}_{\geq 0}$ s.t. $W \cong k^d$.

We will also say W is
finite dimensional

Pf of Proposition ①

$$U \leq k^n$$

WTS: $U \cong k^m$ for $m \leq n$

Pf by induction on n

Base case ($n=0$) ✓

Inductive hypothesis: $U' \leq k^{n-1}$: subspace

then $U' \cong k^{m'}$, $m' \leq n-1$

Inductive step:

$$k^{n-1} = \{(a_1, \dots, a_{n-1}, 0)\} \leq k^n$$

$$\downarrow$$
$$k^n / k^{n-1}$$

$$\begin{array}{ccc} k^n & \xrightarrow{(a_1, \dots, a_n)} & \\ \downarrow & \searrow p & \nearrow a_n \\ k^n / k^{n-1} & \xrightarrow{\cong} & k \end{array}$$

Two facts:

$$(a) \quad U \cap k^{n-1} \leq k^{n-1}$$

Then by inductive hyp.

$$U \cap k^{n-1} \cong k^{m'}, \quad \exists m' \leq n-1$$

$$(h) \quad p(u) \leq k$$

Then:

$$\{p(v) : v \in u\}$$

(i) either $p(u) = \{0\}$

(ii) or $p(u) = k$

Rmk: The only k -subspaces of k are $\{0\}$ & k . This is because of the existence of inverse

$$0 \neq a \in k \Rightarrow a^{-1} \cdot a = 1$$

$$0 \neq a \in W \leq k \Rightarrow 1 \in W \Rightarrow W = k.$$

Contrast: The subgroups of Z_l are of the form

$$dZ_l \leq Z_l$$

for $d \in Z_l$

We obtain a factoring triangle

$$\begin{array}{ccc} U \cap k^{n-1} \leq U & & \\ \text{sl } k^{m'} \downarrow & \searrow \text{plu: ker} = U \cap k^{n-1} & \\ U/(U \cap k^{n-1}) & \xrightarrow{\cong} & p(U) \leq k \end{array}$$

Then:

$$(i) \text{ either } p(U) = \{0\} \quad \} \Leftrightarrow U/(U \cap k^{n-1}) = 0$$

$$\Leftrightarrow U \cap k^{n-1} = U$$

$$\Rightarrow U \cong k^{m'}, \quad \boxed{m' \leq n-1 < n}$$

$$(ii) \text{ or } p(U) = k \quad \}$$

$$\begin{array}{ccc} U \cap k^{n-1} \leq U & & \\ \text{sl } k^{m'} \downarrow & & \\ \underline{k^{m'}} & & U/(U \cap k^{n-1}) \cong \underline{k} \end{array}$$

Claim: $U \cong k^{n'+1} = k^{m'} \times k$

Pf: Choose $v \in U$ s.t.

$p(v) = 1 \in k$ (i.e. last co-ordinate of v is non-zero)

$$V = \langle v \rangle_k = \{a \cdot v : a \in k\} \\ \leq U$$

$$p|_V : V \longrightarrow k \\ a \cdot v \longmapsto a \cdot 1 = a$$

is an isomorphism

This gives a Complement to $k^{n-1} \cap U \leq U$ and shows that

$$U \cong (k^{n-1} \cap U) \times V$$

$$\cong k^{m'} \times k$$

$$\cong k^{m'+1}$$

$$\boxed{m'+1 \leq n-1+1 = n} \quad \checkmark$$

